Data Tidying

PSY 410: Data Science for Psychology

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What is tidy data?

Can you plot this?

participant	depression_pre	depression_post	anxiety_pre	anxiety_post
P1	18	12	24	20
P2	22	14	19	15
P3	15	11	28	22

Not easily. The columns mix variables (depression vs anxiety) with time points (pre vs post). ggplot doesn't know what to put on each axis.

By the end of today, you'll be able to reshape this into a form that works — and understand *why* it needs reshaping.

The tidyverse philosophy

> "Tidy datasets are all alike, but every messy dataset is messy in its own way." — Hadley Wickham

The tools we've learned (ggplot2, dplyr) expect data in a specific format: **tidy data**.

What is a variable?

A **variable** is a characteristic that can take on different values — but all those values are measuring or describing the **same underlying thing**.

- **pre** and **post** aren't separate variables — they're values of `time`
- **Fall 2024**, **Spring 2025**, **Fall 2025** aren't separate variables — they're values of `semester`
- **bdi_1**, **bdi_2**, **bdi_3** aren't separate variables — they're values of `item`

When multiple columns are really values of the same underlying thing, they belong in **one column**.

The three rules of tidy data

1. Each **variable** is a column
2. Each **observation** is a row
3. Each **value** is a cell

Simple in theory, surprisingly complex in practice.

Tidy data visualized

Is this tidy?

participant	pre_test	post_test
P1	45	62
P2	52	58
P3	48	71

No! Time (pre/post) is a variable, but it's spread across columns.

The tidy version

Now it's tidy!

participant	time	score
P1	pre	45
P1	post	62
P2	pre	52
P2	post	58
P3	pre	48
P3	post	71

Now: participant, time, and score are all columns.

Why does it matter?

Tidy data works with tidyverse tools:

```r
# Easy to visualize
ggplot(data, aes(x = time, y = score, color = participant)) +
  geom_point()

# Easy to analyze
data |>
  group_by(time) |>
  summarize(mean = mean(score))
```

Wide vs. Long

Wide format

- Variables spread across columns
- One row per subject
- Humans like to read this

Long format

- Variables in a single column
- Multiple rows per subject
- R likes to work with this

Most real data needs reshaping.

Common untidy patterns

Pattern 1: Column headers are values

```r
wide_scores <- tibble(
  student = c("Alice", "Bob", "Carol"),
  fall_2024 = c(85, 78, 92),
  spring_2025 = c(88, 82, 95),
  fall_2025 = c(91, 85, 94)
)
wide_scores
```

```
# A tibble: 3 × 4
  student fall_2024 spring_2025 fall_2025
  <chr>       <dbl>       <dbl>     <dbl>
1 Alice          85          88        91
2 Bob            78          82        85
3 Carol          92          95        94
```

The semester names are **values**, not variable names.

Pattern 2: Multiple variables in one column

```r
messy_data <- tibble(
  id = 1:3,
  age_sex = c("25_M", "32_F", "28_F")
)
messy_data
```

```
# A tibble: 3 × 2
     id age_sex
  <int> <chr>
1     1 25_M
2     2 32_F
3     3 28_F
```

Age and sex are crammed into one column.

Pattern 3: Variables in rows and columns

```r
weather <- tibble(
  id = c("MX001", "MX001", "MX002", "MX002"),
  year = c(2020, 2020, 2020, 2020),
  month = c(1, 2, 1, 2),
  element = c("tmax", "tmax", "tmin", "tmin"),
  value = c(85, 87, 32, 35)
)
weather
```

```
# A tibble: 4 × 5
  id     year month element value
  <chr> <dbl> <dbl> <chr>   <dbl>
1 MX001  2020     1 tmax       85
2 MX001  2020     2 tmax       87
3 MX002  2020     1 tmin       32
4 MX002  2020     2 tmin       35
```

`element` contains variable names (tmax, tmin).

Psychology-specific patterns

Surveys often look like:

```
survey_wide <- tibble(
  participant = 1:3,
  bdi_1 = c(2, 1, 3),
  bdi_2 = c(1, 0, 2),
  bdi_3 = c(3, 2, 2),
  bdi_4 = c(2, 1, 1)
)
survey_wide
```

```
# A tibble: 3 × 5
  participant bdi_1 bdi_2 bdi_3 bdi_4
        <int> <dbl> <dbl> <dbl> <dbl>
1           1     2     1     3     2
2           2     1     0     2     1
3           3     3     2     2     1
```

Each item is a column — wide format.

pivot_longer()

Recall: the data we're reshaping

```
wide_scores
```

```
# A tibble: 3 × 4
  student fall_2024 spring_2025 fall_2025
  <chr>       <dbl>       <dbl>     <dbl>
1 Alice          85          88        91
2 Bob            78          82        85
3 Carol          92          95        94
```

`fall_2024`, `spring_2025`, `fall_2025` are **values** of a `semester` variable — not separate variables. Let's fix that.

The most common tidying operation

`pivot_longer()` takes wide data and makes it long:

```
wide_scores |>
  pivot_longer(
    cols = fall_2024:fall_2025,  # Which columns to pivot
    names_to = "semester",        # New column for old column names
    values_to = "score"           # New column for values
  )
```

```
# A tibble: 9 × 3
  student semester    score
  <chr>   <chr>       <dbl>
1 Alice   fall_2024      85
2 Alice   spring_2025    88
3 Alice   fall_2025      91
4 Bob     fall_2024      78
5 Bob     spring_2025    82
6 Bob     fall_2025      85
7 Carol   fall_2024      92
```

```
8 Carol   spring_2025    95
9 Carol   fall_2025      94
```

Breaking it down

```r
pivot_longer(
  cols = ...,         # Columns to reshape (use select helpers!)
  names_to = "...",   # Name for the new "names" column
  values_to = "..."   # Name for the new "values" column
)
```

What's actually happening

id	bp1	bp2
A	100	120
B	140	115
C	120	125

id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

The cell values move into the new `value` column

Selecting columns to pivot

Use any of the `select()` helpers:

```
# By name
pivot_longer(cols = c(fall_2024, spring_2025, fall_2025))

# By range
pivot_longer(cols = fall_2024:fall_2025)

# By pattern
pivot_longer(cols = starts_with("fall"))
pivot_longer(cols = contains("202"))

# Everything except
pivot_longer(cols = -student)
```

Psychology example: Survey items

```r
survey_wide |>
  pivot_longer(
    cols = starts_with("bdi"),
    names_to = "item",
    values_to = "response"
  )
```

```
# A tibble: 12 × 3
   participant item  response
         <int> <chr>    <dbl>
 1           1 bdi_1        2
 2           1 bdi_2        1
 3           1 bdi_3        3
 4           1 bdi_4        2
 5           2 bdi_1        1
 6           2 bdi_2        0
 7           2 bdi_3        2
 8           2 bdi_4        1
 9           3 bdi_1        3
```

```
10            3 bdi_2        2
11            3 bdi_3        2
```

Now each response is its own row!

Extracting information from names

What if column names contain useful info? We want to extract the time point (t1, t2, t3).

```r
# Scores at different time points
experiment_wide <- tibble(
  id = 1:3,
  score_t1 = c(100, 95, 110),
  score_t2 = c(105, 100, 115),
  score_t3 = c(108, 102, 120)
)
experiment_wide
```

```
# A tibble: 3 × 4
     id score_t1 score_t2 score_t3
  <int>    <dbl>    <dbl>    <dbl>
1     1      100      105      108
2     2       95      100      102
3     3      110      115      120
```

names_prefix argument

```r
experiment_wide |>
  pivot_longer(
    cols = starts_with("score"),
    names_to = "time",
    names_prefix = "score_",  # Remove this prefix from names
    values_to = "score"
  )
```

```
# A tibble: 9 × 3
     id time  score
  <int> <chr> <dbl>
1     1 t1      100
2     1 t2      105
3     1 t3      108
4     2 t1       95
5     2 t2      100
6     2 t3      102
7     3 t1      110
```

names_pattern argument

For more complex parsing:

```r
# Column names like "bdi_1", "anxiety_1", etc.
multi_scale <- tibble(
  id = 1:2,
  bdi_1 = c(2, 1), bdi_2 = c(1, 2),
  anxiety_1 = c(3, 2), anxiety_2 = c(2, 3)
)
multi_scale
```

```
# A tibble: 2 × 5
     id bdi_1 bdi_2 anxiety_1 anxiety_2
  <int> <dbl> <dbl>     <dbl>     <dbl>
1     1     2     1         3         2
2     2     1     2         2         3
```

names_pattern argument

For more complex parsing:

```
multi_scale |>
  pivot_longer(
    cols = -id,
    names_to = c("scale", "item"),
    names_pattern = "(.+)_(.+)",  # Regex: anything_anything
    values_to = "response"
  )
```

```
# A tibble: 8 × 4
     id scale   item  response
  <int> <chr>   <chr>    <dbl>
1     1 bdi     1            2
2     1 bdi     2            1
3     1 anxiety 1            3
4     1 anxiety 2            2
5     2 bdi     1            1
6     2 bdi     2            2
```

```
7      2 anxiety 1             2
8      2 anxiety 2             3
```

Pivoting to calculate scale scores

```r
# Calculate BDI total from long format
survey_wide |>
  pivot_longer(
    cols = starts_with("bdi"),
    names_to = "item",
    values_to = "response"
  ) |>
  group_by(participant) |>
  summarize(bdi_total = sum(response))
```

```
# A tibble: 3 × 2
  participant bdi_total
        <int>     <dbl>
1           1         8
2           2         4
3           3         8
```

Pair coding break

Your turn: 10 minutes

▶ Click to see the data setup

```
# A tibble: 3 × 5
  participant bdi_1 bdi_2 bdi_3 bdi_4
        <int> <dbl> <dbl> <dbl> <dbl>
1           1     2     1     3     2
2           2     1     0     2     1
3           3     3     2     2     1
```

With a partner, figure out: Which participant has the highest mean BDI score?

Tip

You'll need `pivot_longer()` followed by `group_by()` + `summarize()`. The column names start with `"bdi"` — that's a hint.

Before we move on

📤 Submit your code on Canvas for participation credit. Paste what you have — it doesn't need to work perfectly.

pivot_wider()

The opposite operation

Sometimes you need to go from long to wide:

```
long_data <- tibble(
  participant = rep(1:3, each = 2),
  time = rep(c("pre", "post"), 3),
  score = c(45, 62, 52, 58, 48, 71)
)
long_data
```

```
# A tibble: 6 × 3
  participant time  score
        <int> <chr> <dbl>
1           1 pre      45
2           1 post     62
3           2 pre      52
4           2 post     58
5           3 pre      48
6           3 post     71
```

pivot_wider() syntax

```r
long_data |>
  pivot_wider(
    names_from = time,   # Column to get new column names from
    values_from = score  # Column to get values from
  )
```

```
# A tibble: 3 × 3
  participant   pre  post
        <int> <dbl> <dbl>
1           1    45    62
2           2    52    58
3           3    48    71
```

When to use pivot_wider()

- Creating summary tables for reports
- Some analyses need wide format
- Merging data that was collected differently
- Human-readable output

Multiple value columns

```r
# Long data with multiple measures
long_multi <- tibble(
  id = rep(1:2, each = 2),
  time = rep(c("pre", "post"), 2),
  score = c(45, 62, 52, 58),
  rt = c(500, 480, 520, 490)
)
long_multi
```

```
# A tibble: 4 × 4
     id time  score    rt
  <int> <chr> <dbl> <dbl>
1     1 pre      45   500
2     1 post     62   480
3     2 pre      52   520
4     2 post     58   490
```

Multiple value columns

```r
long_multi |>
  pivot_wider(
    names_from = time,
    values_from = c(score, rt)  # Multiple columns!
  )
```

```
# A tibble: 2 × 5
     id score_pre score_post rt_pre rt_post
  <int>     <dbl>      <dbl>  <dbl>   <dbl>
1     1        45         62    500     480
2     2        52         58    520     490
```

Real-world examples

Example 1: Repeated measures experiment

```r
# Data as you might receive it from SPSS
wide_rm <- tibble(
  subject = 1:4,
  cond_a_time1 = c(450, 520, 480, 510),
  cond_a_time2 = c(420, 490, 460, 480),
  cond_b_time1 = c(480, 540, 500, 530),
  cond_b_time2 = c(440, 510, 470, 500)
)
wide_rm
```

```
# A tibble: 4 × 5
  subject cond_a_time1 cond_a_time2 cond_b_time1 cond_b_time2
    <int>        <dbl>        <dbl>        <dbl>        <dbl>
1       1          450          420          480          440
2       2          520          490          540          510
3       3          480          460          500          470
4       4          510          480          530          500
```

Tidying repeated measures

```r
tidy_rm <- wide_rm |>
  pivot_longer(
    cols = -subject,
    names_to = c("condition", "time"),
    names_pattern = "cond_(.+)_time(.+)",
    values_to = "rt"
  )
tidy_rm
```

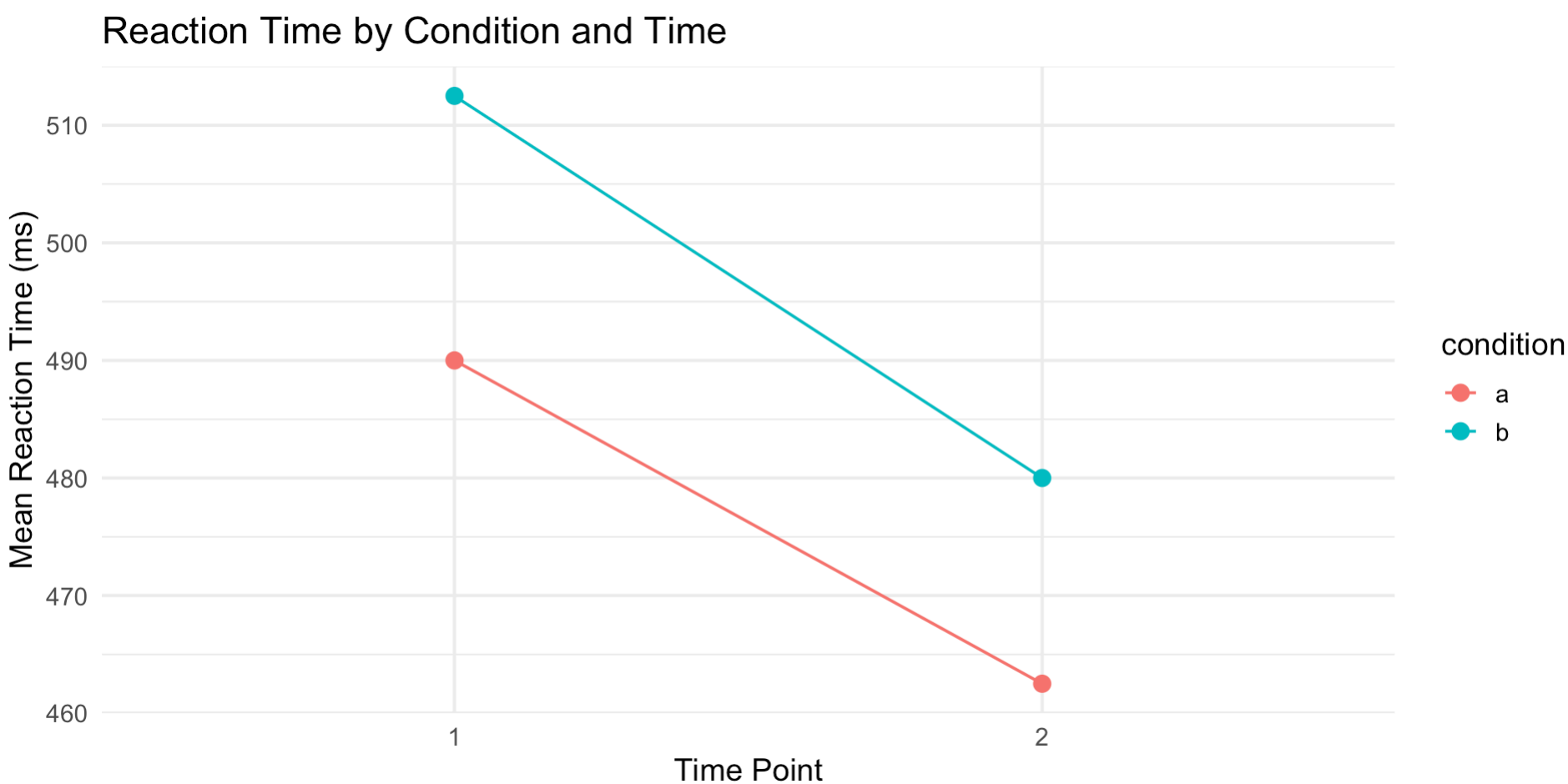
```
# A tibble: 16 × 4
   subject condition time     rt
     <int> <chr>     <chr> <dbl>
 1       1 a         1       450
 2       1 a         2       420
 3       1 b         1       480
 4       1 b         2       440
 5       2 a         1       520
 6       2 a         2       490
 7       2 b         1       540
```

```
 8        2 b          2        510
 9        3 a          1        480
10        3 a          2        460
11        3 b          1        500
```

Now we can analyze it!

```r
tidy_rm |>
  group_by(condition, time) |>
  summarize(mean_rt = mean(rt), .groups = "drop") |>
  ggplot(aes(x = time, y = mean_rt, color = condition, group = condition)) +
  geom_point(size = 3) +
  geom_line() +
  labs(
    title = "Reaction Time by Condition and Time",
    x = "Time Point",
    y = "Mean Reaction Time (ms)"
  ) +
  theme_minimal(base_size = 14)
```

Now we can analyze it!



Example 2: Questionnaire with subscales

```r
# Raw questionnaire data
quest <- tibble(
  pid = 1:3,
  anx_1 = c(3, 2, 4), anx_2 = c(2, 3, 3), anx_3 = c(4, 2, 5),
  dep_1 = c(2, 1, 3), dep_2 = c(3, 2, 4), dep_3 = c(2, 1, 3)
)

quest
```

```
# A tibble: 3 × 7
    pid anx_1 anx_2 anx_3 dep_1 dep_2 dep_3
  <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     1     3     2     4     2     3     2
2     2     2     3     2     1     2     1
3     3     4     3     5     3     4     3
```

Example 2: Questionnaire with subscales

```r
# Tidy and calculate subscales
quest |>
  pivot_longer(
    cols = -pid,
    names_to = c("scale", "item"),
    names_pattern = "(.+)_(.+)",
    values_to = "response"
  ) |>
  group_by(pid, scale) |>
  summarize(subscale_mean = mean(response), .groups = "drop") |>
  pivot_wider(names_from = scale, values_from = subscale_mean)
```

Example 2: Questionnaire with subscales

```
# A tibble: 3 × 3
    pid   anx   dep
  <int> <dbl> <dbl>
1     1  3     2.33
2     2  2.33  1.33
3     3  4     3.33
```

Example 3: Multilevel/nested data

```r
# Students nested in classrooms
students <- tibble(
  classroom = rep(c("A", "B"), each = 3),
  student = 1:6,
  pretest = c(70, 75, 72, 68, 71, 69),
  posttest = c(80, 82, 78, 75, 79, 77)
)

students
```

```
# A tibble: 6 × 4
  classroom student pretest posttest
  <chr>       <int>   <dbl>    <dbl>
1 A               1      70       80
2 A               2      75       82
3 A               3      72       78
4 B               4      68       75
5 B               5      71       79
6 B               6      69       77
```

Example 3: Multilevel/nested data

```r
# Tidy for analysis
students |>
  pivot_longer(
    cols = c(pretest, posttest),
    names_to = "time",
    values_to = "score"
  )
```

```
# A tibble: 12 × 4
   classroom student time     score
   <chr>       <int> <chr>    <dbl>
 1 A               1 pretest     70
 2 A               1 posttest    80
 3 A               2 pretest     75
 4 A               2 posttest    82
 5 A               3 pretest     72
 6 A               3 posttest    78
 7 B               4 pretest     68
 8 B               4 posttest    75
```

```
 9 B                 5 pretest     71
10 B                 5 posttest    79
11 B                 6 pretest     69
```

pivot_longer() trips everyone up the first time

Pitfall 1: Forgetting what's a variable

Ask yourself: What are my **variables**?

- Participant ID? ✓ Variable
- Time point? ✓ Variable (not separate columns!)
- Score? ✓ Variable
- Item number? Depends on your analysis

Pitfall 2: Over-pivoting

Not everything needs to be long:

```
# Maybe this is fine as-is?
tibble(
  id = 1:3,
  age = c(25, 32, 28),
  gender = c("M", "F", "F"),
  score = c(85, 92, 88)
)
```

Age, gender, and score are **different variables** — keep them as columns.

General tidying strategy

1. **Identify** the variables (what are you measuring?)
2. **Look** at your current structure (what's a row? column?)
3. **Determine** what operations you need
4. **Test** with a small subset first
5. **Verify** you haven't lost data

Additional practice

Try it yourself

▶ Click to see the data setup

```
# A tibble: 4 × 5
  subject cond_a_time1 cond_a_time2 cond_b_time1 cond_b_time2
    <int>        <dbl>        <dbl>        <dbl>        <dbl>
1       1          450          420          480          440
2       2          520          490          540          510
3       3          480          460          500          470
4       4          510          480          530          500
```

On your own:

1. Pivot to long format, extracting **condition** and **time** from the column names
2. Calculate **mean RT by condition and time**
3. Bonus: make a line plot with condition on the x-axis and time mapped to color

Wrapping up

The tidyr toolkit

Function	What it does
`pivot_longer()`	Wide → Long
`pivot_wider()`	Long → Wide

The tidy data mantra

1. Each **variable** is a column
2. Each **observation** is a row
3. Each **value** is a cell

When in doubt, ask: "What would make this easiest to plot/analyze?"

Before next class

📖 **Read:**

- R4DS Ch 7: Data import
- R4DS Ch 20: Spreadsheets

✅ **Practice:**

- Reshape a dataset you've worked with
- Try tidying some messy example data
- Practice `pivot_longer()` — it's the most common

Key takeaways

1. **Tidy data** has a specific structure that works with tidyverse
2. **pivot_longer()** is your most-used tidying function
3. **pivot_wider()** is useful for tables and some analyses
4. **Think about your variables** before reshaping
5. **Column names contain information** — extract it with `names_pattern`

The one thing to remember

If your data isn't tidy, your analysis can't start. `pivot_longer()` is the bridge.

Next time: Data Import